TinyHAR: A Lightweight Deep Learning Model for Human Activity Recognition

Department of Computer Science & Artificial Intelligence

Abstract

This paper investigates activity recognition with extensive empirical and theoretical analysis. We evaluate novel architectures, optimization protocols, and benchmark datasets. Our findings demonstrate competitive state-of-the-art results across standard scientific benchmarks.

1. Introduction

Recent advancements in machine learning have emphasized the importance of rigorous methodological foundations. In this work, we present systematic findings regarding TinyHAR: A Lightweight Deep Learning Model for Human Activity Recognition.

Section topic: Activity Recognition
Question context: Why is battery and computational efficiency critical for edge HAR on microcontrollers?
Key Finding: Edge wearables feature microcontrollers with strict memory constraints and tiny batteries, demanding lightweight quantized models that avoid heavy recurrent activations [TinyHAR, p. 2].
Evidence passage: Wearable smartwatches and IoT nodes have constrained RAM (under 512 KB) and battery life Complex recurrent models cause thermal throttling and fast battery drain

2. Technical Methodology & Empirical Findings

Continuing detailed technical derivations for TinyHAR: A Lightweight Deep Learning Model for Human Activity Recognition. We observe consistent experimental stability across multi-trial evaluations. The parameters exhibit robust convergence with bounded variance. Ablation studies confirm the necessity of the proposed design components under rigorous baseline comparisons.

3. Technical Methodology & Empirical Findings

Section topic: Multi-Paper Analysis
Question context: How do DeepConvLSTM and TinyHAR compare in computational overhead for wearable deployment?
Key Finding: DeepConvLSTM incurs high memory overhead from recurrent cell state propagation [DeepConvLSTM, p. 3], whereas TinyHAR replaces heavy layers with depthwise convolutions for edge microcontrollers [TinyHAR, p. 4].
Evidence passage: DeepConvLSTM requires sequential recurrence and substantial memory for hidden states TinyHAR uses depthwise separable convolutions to cut parameter count by 80%

4. Technical Methodology & Empirical Findings

Section topic: Activity Recognition
Question context: How does 8-bit integer quantization (INT8) affect HAR model accuracy and latency?
Key Finding: INT8 post-training quantization cuts model size by 75% and speeds up inference by 2-3x while maintaining within 1% of full-precision F1 accuracy [TinyHAR, p. 5].
Evidence passage: Reduces model memory footprint by 4x Decreases inference latency by 2-3x with less than 1% degradation in F1-score

5. Technical Methodology & Empirical Findings

Continuing detailed technical derivations for TinyHAR: A Lightweight Deep Learning Model for Human Activity Recognition. We observe consistent experimental stability across multi-trial evaluations. The parameters exhibit robust convergence with bounded variance. Ablation studies confirm the necessity of the proposed design components under rigorous baseline comparisons.